

Teorema de la divergencia de Gauss

Sea $\mathbf{F} := [F_1, F_2]^T \in [C^1(\overline{\Omega})]^2$, con Ω un dominio acotado. Entonces:

$$\int_{\Omega} \operatorname{div} \mathbf{F} dA = \int_{\partial\Omega} \mathbf{F} \cdot \mathbf{n} d\ell$$

donde $\mathbf{n} = [n_1, n_2]^T$ es vector normal unitario exterior a $\partial\Omega$.

En particular si $\mathbf{F} = \begin{bmatrix} uv \\ 0 \end{bmatrix}$ y $\mathbf{F} = \begin{bmatrix} 0 \\ uv \end{bmatrix}$ donde $u, v \in C^1(\Omega) \cap C(\overline{\Omega})$, entonces

$$\int_{\Omega} (uv)_{x_i} dA = \int_{\partial\Omega} uv n_i d\ell \quad \rightsquigarrow \quad \int_{\Omega} uv_{x_i} dA + \int_{\Omega} u_{x_i} v dA = \int_{\partial\Omega} uv n_i d\ell$$

ahora, si $v = w_{x_i}$ con $w \in C^2(\Omega) \cap C^1(\overline{\Omega})$

$$\int_{\Omega} uw_{x_i x_i} dA + \int_{\Omega} u_{x_i} w_{x_i} dA = \int_{\partial\Omega} uw_{x_i} n_i d\ell$$

y sumamos para i de 1 a 2, obtenemos

Identidad de Green

Si $u \in C^1(\Omega) \cap C(\overline{\Omega})$ y $w \in C^2(\Omega) \cap C^1(\overline{\Omega})$

$$\int_{\Omega} u \Delta w + \int_{\Omega} \nabla u \cdot \nabla w = \int_{\partial\Omega} u \nabla w \cdot \mathbf{n}$$

donde $\nabla u = [u_{x_1}, u_{x_2}]^T$.

Derivada parcial débil

Si $v \in L^2(\Omega)$ tiene una derivada débil (o en el sentido de las distribuciones) en la dirección x_i si

$$\int_{\Omega} v \varphi_{x_i} = - \int_{\Omega} w \varphi \quad \forall \varphi \in C_0^\infty(\Omega).$$

$$H^1(\Omega) := \{v \in L^2(\Omega) : \nabla v \in [L^2(\Omega)]^2\}$$

$$(u, v)_{H^1(\Omega)} := (u, v)_{L^2(\Omega)} + \sum_{i=1}^2 (u_{x_i}, v_{x_i})_{L^2(\Omega)} = (u, v)_{L^2(\Omega)} + (\nabla u, \nabla v)_{L^2(\Omega)}$$

Denotaremos la **seminorma** en $H^1(\Omega)$ por $|v|_{H^1(\Omega)}^2 := (\nabla v, \nabla v)_{L^2(\Omega)}$.

Norma en $H^1(\Omega)$

$$\|v\|_{H^1(\Omega)}^2 = \|v\|_{L^2(\Omega)}^2 + |v|_{H^1(\Omega)}^2.$$

Espacio Hilbert $(L^2(\Gamma), (\cdot, \cdot)_{L^2(\Gamma)})$ para $\Gamma := \partial\Omega$

Definimos el E.V.

$$L^2(\Gamma) := \left\{ w : \Gamma \rightarrow \mathbb{R} : w \text{ es medible y } \int_{\Gamma} w^2 < \infty \right\}$$

con el producto interno

$$(w, z)_{L^2(\Gamma)} := \int_{\Gamma} wz$$

Teorema de Trazas

Sea Ω un abierto 1-regular, la aplicación $\gamma_0 : H^1(\Omega) \rightarrow L^2(\Gamma)$ definida por $\gamma_0(v) := v|_{\Gamma}$ es lineal y continua, es decir, existe $C > 0$ tal que

$$\|\gamma_0(v)\|_{L^2(\Gamma)} = \|v|_{\Gamma}\|_{L^2(\Gamma)} = \|v\|_{L^2(\Gamma)} \leq C \|u\|_{H^1(\Omega)}.$$

Desigualdad Generalizada de Poincaré

Sean Ω un abierto 1-regular y $\mathcal{F}$ una familia de funcionales lineales y acotados definidos en $H^m(\Omega)$ ($m > 0$). Si se cumple es tal que

$$(v \in \mathbb{P}_{m-1}(\Omega) \text{ es tal que } F(v) = 0 \quad \forall F \in \mathcal{F}) \quad \Rightarrow \quad v \equiv 0$$

entonces la aplicación

$$\|\cdot\| : H^m(\Omega) \longrightarrow \mathbb{R}$$

$$v \longmapsto \|v\| := \left(|v|_{H^m(\Omega)}^2 + \sum_{F \in \mathcal{F}} |F(v)|^2 \right)^{1/2}$$

define una norma equivalente a $H^m(\Omega)$.

Aquí, $\mathbb{P}_k(\Omega)$ es el conjunto de polinomios de grado $\leq k$ definidos en Ω .

Ejemplos

Sea Ω un abierto 1-regular de $H^1(\Omega)$ y $\Gamma := \Gamma_D \cup \Gamma_N$

$$H_0^1(\Omega) := \{v \in H^1(\Omega) : v = 0 \text{ sobre } \Gamma\}$$

$$\begin{aligned} F : H^1(\Omega) &\longrightarrow \mathbb{R} \\ v &\longmapsto F(v) := \int_{\Gamma} v \end{aligned}$$

$$H_{\Gamma_D}^1(\Omega) := \{v \in H^1(\Omega) : v = 0 \text{ sobre } \Gamma_D\}$$

$$\begin{aligned} G : H^1(\Omega) &\longrightarrow \mathbb{R} \\ v &\longmapsto G(v) := \int_{\Gamma_D} v \end{aligned}$$

$$H_{\text{prom}}^1(\Omega) := \{v \in H^1(\Omega) : \int_{\Omega} v = 0\}$$

$$\begin{aligned} K : H^1(\Omega) &\longrightarrow \mathbb{R} \\ v &\longmapsto K(v) := \int_{\Omega} v \end{aligned}$$

E.D.P. Elíptica: sea $\Omega \subseteq \mathbb{R}^2$ y $\Gamma := \Gamma_D \cup \Gamma_N$

Queremos aproximar u solución del problema:

$$\begin{array}{ll} \text{EH} & \left\{ \begin{array}{ll} -\Delta u = f, & \text{en } \Omega, \\ u = 0, & \text{sobre } \Gamma. \end{array} \right. & \text{EM} & \left\{ \begin{array}{ll} -\Delta u = f, & \text{en } \Omega, \\ u = 0, & \text{sobre } \Gamma_D, \\ \nabla u \cdot \mathbf{n} = g, & \text{sobre } \Gamma_N, \end{array} \right. \end{array}$$

$$\begin{array}{ll} \text{EPN} & \left\{ \begin{array}{ll} -\Delta u = f, & \text{en } \Omega, \\ \nabla u \cdot \mathbf{n} = g, & \text{sobre } \Gamma, \\ \int_{\Omega} v = 0 \end{array} \right. & \text{EMNH} & \left\{ \begin{array}{ll} -\Delta u = f, & \text{en } \Omega, \\ u = g_1, & \text{sobre } \Gamma_D, \\ \nabla u \cdot \mathbf{n} = g_2, & \text{sobre } \Gamma_N, \end{array} \right. \end{array}$$

$$\text{E} \quad \left\{ \begin{array}{ll} u - \Delta u = f, & \text{en } \Omega, \\ \nabla u \cdot \mathbf{n} = g, & \text{sobre } \Gamma. \end{array} \right.$$

Aproximación por elementos finitos

Supongamos que $\Omega \subset \mathbb{R}^2$ y Γ es poligonal.

Definición: $\mathcal{T}_h$ es una triangulación de $\bar{\Omega}$ si

- 1 $\bigcup_{i=1}^p T_i = \bar{\Omega}$ (i.e. son una partición de $\bar{\Omega}$),
- 2 para $T_i, T_j \in \mathcal{T}_h$ ocurre solo uno de los siguientes casos:
 - $T_i \cap T_j = \emptyset$
 - $T_i \cap T_j = \text{lado}$ (i.e. tiene un lado en común)
 - $T_i \cap T_j = \text{vértice}$ (i.e. tiene un vértice en común)

Subespacios de dimensión finita de $H_0^1(\Omega)$ y $H_{\Gamma_D}^1(\Omega)$

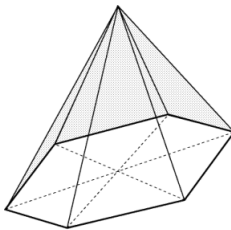
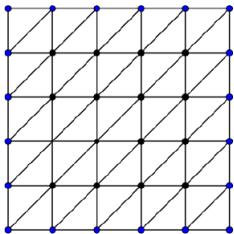
$$V_h^\Gamma := \{v \in C(\bar{\Omega}) : v|_T \in \mathbb{P}_1(T), \quad \forall T \in \mathcal{T}_h, \quad v = 0 \text{ sobre } \Gamma\},$$

$$V_h^{\Gamma_D} := \{v \in C(\bar{\Omega}) : v|_T \in \mathbb{P}_1(T), \quad \forall T \in \mathcal{T}_h, \quad v = 0 \text{ sobre } \Gamma_D\}.$$

Base de los espacios finito-dimensionales

Sean $\{\mathbf{x}_m\}_{m=1}^r$ el conjunto de todos los vértices (nodos) de $\mathcal{T}_h$, entonces podemos denotar:

$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_s \quad \longleftarrow \quad \text{nodos interiores}$
 $\mathbf{x}_{s+1}, \mathbf{x}_{s+2}, \dots, \mathbf{x}_r \quad \longleftarrow \quad \text{nodos de frontera}$



$$u_h = \sum_{m=1}^{\star} \alpha_m \varphi_m$$

$$u_h|_T = \alpha_1 \varphi_1 + \alpha_2 \varphi_2 + \alpha_3 \varphi_3 \quad \longleftarrow \quad \textit{localmente}$$

$$\varphi_m(x, y) = a_m + b_m x + c_m y$$

$$\varphi_m(\mathbf{x}_j) = \delta_{j,m}$$

$$\nabla \varphi_m(\mathbf{x}) = \begin{bmatrix} b_m \\ c_m \end{bmatrix}$$

$$\varphi_1 : \begin{cases} a_1 + b_1 x_1 + c_1 y_1 = 1 \\ a_1 + b_1 x_2 + c_1 y_2 = 0 \\ a_1 + b_1 x_3 + c_1 y_3 = 0 \end{cases}$$

$$\varphi_2 : \begin{cases} a_2 + b_2 x_1 + c_2 y_1 = 0 \\ a_2 + b_2 x_2 + c_2 y_2 = 1 \\ a_2 + b_2 x_3 + c_2 y_3 = 0 \end{cases}$$

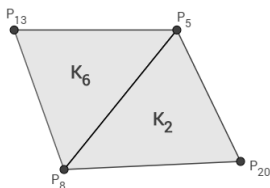
$$\varphi_3 : \begin{cases} a_3 + b_3 x_1 + c_3 y_1 = 0 \\ a_3 + b_3 x_2 + c_3 y_2 = 0 \\ a_3 + b_3 x_3 + c_3 y_3 = 1 \end{cases}$$

$$\sim \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{bmatrix} = \mathbf{I}$$

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Si denotamos $\mathbf{G} := \begin{bmatrix} b_1 & c_1 \\ b_2 & c_2 \\ b_3 & c_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{bmatrix}^{-1} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$, tenemos

$$\begin{aligned} \int_T \nabla u_h \cdot \nabla v_h &= \nabla u_h|_T \cdot \nabla v_h|_T \int_T dA \\ &= |T| \nabla(\alpha_1 \varphi_1 + \alpha_2 \varphi_2 + \alpha_3 \varphi_3) \cdot \nabla(\beta_1 \varphi_1 + \beta_2 \varphi_2 + \beta_3 \varphi_3) \\ &= |T| \begin{bmatrix} \beta_1 & \beta_2 & \beta_3 \end{bmatrix} \begin{bmatrix} \nabla \varphi_1 \cdot \nabla \varphi_1 & \nabla \varphi_2 \cdot \nabla \varphi_1 & \nabla \varphi_3 \cdot \nabla \varphi_1 \\ \nabla \varphi_1 \cdot \nabla \varphi_2 & \nabla \varphi_2 \cdot \nabla \varphi_2 & \nabla \varphi_3 \cdot \nabla \varphi_2 \\ \nabla \varphi_1 \cdot \nabla \varphi_3 & \nabla \varphi_2 \cdot \nabla \varphi_3 & \nabla \varphi_3 \cdot \nabla \varphi_3 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \\ &= \begin{bmatrix} \beta_1 & \beta_2 & \beta_3 \end{bmatrix} |T| \mathbf{G} \mathbf{G}^T \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \end{aligned}$$



$$u_h|_{K_2} = \alpha_8 \varphi_8 + \alpha_{20} \varphi_{20} + \alpha_5 \varphi_5$$

$$u_h|_{K_6} = \alpha_5 \varphi_5 + \alpha_{13} \varphi_{13} + \alpha_8 \varphi_8$$

$$A_{K_2} = |K_2| \begin{bmatrix} \nabla \varphi_8 \cdot \nabla \varphi_8 & \nabla \varphi_{20} \cdot \nabla \varphi_8 & \nabla \varphi_5 \cdot \nabla \varphi_8 \\ \nabla \varphi_8 \cdot \nabla \varphi_{20} & \nabla \varphi_{20} \cdot \nabla \varphi_{20} & \nabla \varphi_5 \cdot \nabla \varphi_{20} \\ \nabla \varphi_8 \cdot \nabla \varphi_5 & \nabla \varphi_{20} \cdot \nabla \varphi_5 & \nabla \varphi_5 \cdot \nabla \varphi_5 \end{bmatrix}$$

$$A_{K_6} = |K_6| \begin{bmatrix} \nabla \varphi_5 \cdot \nabla \varphi_5 & \nabla \varphi_{13} \cdot \nabla \varphi_5 & \nabla \varphi_8 \cdot \nabla \varphi_5 \\ \nabla \varphi_5 \cdot \nabla \varphi_{13} & \nabla \varphi_{13} \cdot \nabla \varphi_{13} & \nabla \varphi_8 \cdot \nabla \varphi_{13} \\ \nabla \varphi_5 \cdot \nabla \varphi_8 & \nabla \varphi_{13} \cdot \nabla \varphi_8 & \nabla \varphi_8 \cdot \nabla \varphi_8 \end{bmatrix}$$

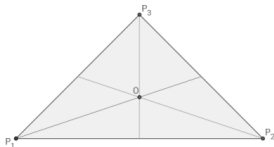
Para construir **A** en MATLAB usamos la instrucción

$$\mathbf{A}([8, 20, 5], [8, 20, 5]) = |K_2| G_{K_2} G_{K_2}^T + \mathbf{A}([8, 20, 5], [8, 20, 5])$$

$$\mathbf{A}([5, 13, 8], [5, 13, 8]) = |K_6| G_{K_6} G_{K_6}^T + \mathbf{A}([5, 13, 8], [5, 13, 8])$$

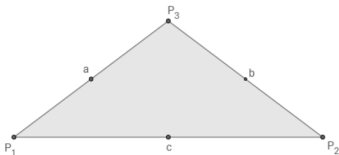
Algunas fórmulas de integración sobre triángulos

- Baricentro: exacta en $\mathbb{P}_1(T)$



$$Q = \frac{1}{3}(p_1 + p_2 + p_3) \quad \longrightarrow \quad \int_T g \approx |T| g(Q)$$

- Punto medio: exacta en $\mathbb{P}_2(T)$



$$Q_1 = \frac{1}{2}(p_1 + p_2), \quad Q_2 = \frac{1}{2}(p_2 + p_3), \quad Q_3 = \frac{1}{2}(p_1 + p_3), \quad \text{pesos} = \frac{1}{3}$$
$$\longrightarrow \quad \int_T g \approx \frac{|T|}{3} (g(Q_1) + g(Q_2) + g(Q_3))$$

GMSH: generador de malla

```
lc = 0.5;  
Point(1) = {0.0,0.0,0.0,lc};  
Point(2) = {2,0.0,0.0,lc};  
Point(3) = {2,2,0.0,lc};  
Point(4) = {0,2,0.0,lc};
```

```
Line(1) = {4,3};  
Line(2) = {3,2};  
Line(3) = {2,1};  
Line(4) = {1,4};
```

```
Line Loop(9) = {2,3,4,1};  
Plane Surface(11) = {9};
```

```
Physical Line(12)={1,2};  
Physical Line(13)={3,4};
```

```
Physical Surface(14)={11};
```